

SMART INDIA HACKATHON 2025



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HACKATHON
2025

- Problem Statement ID – 25050
- Problem Statement Title- SMART TRAFFIC MANAGEMENT SYSTEM
FOR URBAN CONGESTION
- Theme- TRANSPORTATION & LOGISTICS
- PS Category- Software
- Team ID- 92770
- Team Name (Registered on portal) - 404 The Optimists



404- The Optimistics

Lanezy: AI-Powered Adaptive Traffic Control Revolutionizing India's Roads



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Real-world Issue:

Urban congestion, accidents, emergency delays, and poor enforcement cost billions annually, causing massive fuel loss and significant time loss.



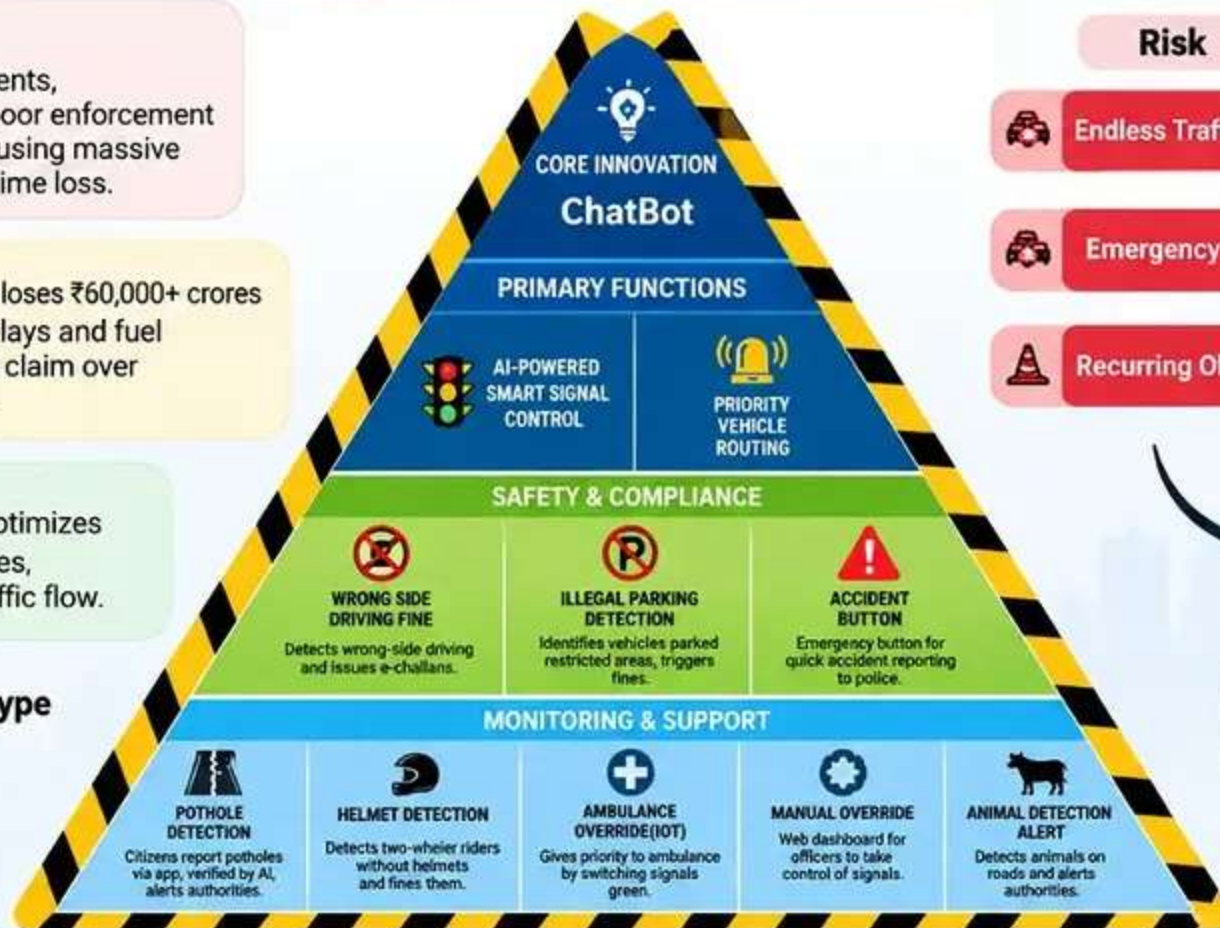
Why important: India loses ₹60,000+ crores annually due to traffic delays and fuel wastage; road accidents claim over 150,000 lives each year.



Solution: AI system optimizes signals, detects anomalies, ensuring safer, faster traffic flow.

Prototype

Video
Website



Risk

VS

Solution



Endless Traffic Jams



Smart Signal Control



Emergency Delays



Priority Vehicle Routing



Recurring Obstacles



Real-time Detection & Alerts



METHODOLOGY & PROCESS OF IMPLEMENTATION

Per-Signal Login

Initiates the control cycle

Parallel Alerts

Responds to incidents and hazards

Emergency Override

Prioritizes emergency vehicle passage

Adaptive Control

Adjusts timings based on density

Manual Override

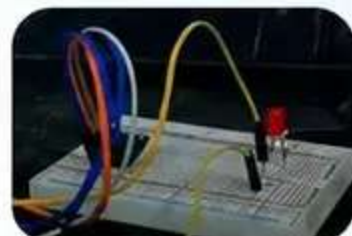
Allows inspector control

Report
GitHub

Detailed Report



SYSTEM IOT



TECHNOLOGIES USED



FrontEnd

React and TailwindCSS are used for the frontend development.



BackEnd

FastAPI is used for the backend development.



Deployment

Render and Vercel are used for deployment.



Hardware

LEDs, GPS Module, ESP8266, YOLOv8.pt, QMC Module, and Resistors are used.



FEASIBILITY AND VIABILITY

PRACTICAL • SCALABLE • IMPACTFUL



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FEASIBILITY ANALYSIS

PRACTICAL TO IMPLEMENT



Infrastructure Reusability

Leverages existing traffic signals and cameras, requiring minimal additional hardware installation.



Scalable Deployment

Can be implemented at a few key intersections initially and gradually expanded citywide as needed.



Cost-Effectiveness

Setup cost is relatively low compared to building entirely new infrastructure.



Authority Integration

Includes features like SOS alerts, obstacle detection, and manual override for seamless police control.



VIABILITY

PROVEN, RELIABLE & TRUSTWORTHY



Proven Concept

Leverages the success of existing AI-driven traffic systems in major cities, demonstrating clear market demand and viability.



Builds Public Trust

Automated detection minimizes human bias, while timestamped video evidence provides credible, transparent proof for all incidents.



Guarantees Accountability

Fosters transparency through detailed logs, acknowledgements, and performance analytics, ensuring every action is traceable.



Ensures Authority Engagement

Real-time dashboards and one-click dispatch alerts keep operators actively engaged and enable rapid response.



BUSINESS POTENTIAL

SUSTAINABLE IMPACT



Government Cost Savings

Automates manual monitoring to significantly reduce operational costs and optimize the allocation of traffic personnel.



New Revenue Generation

Creates a direct and sustainable revenue stream through seamless integration with e-challan systems for automated fine collection.

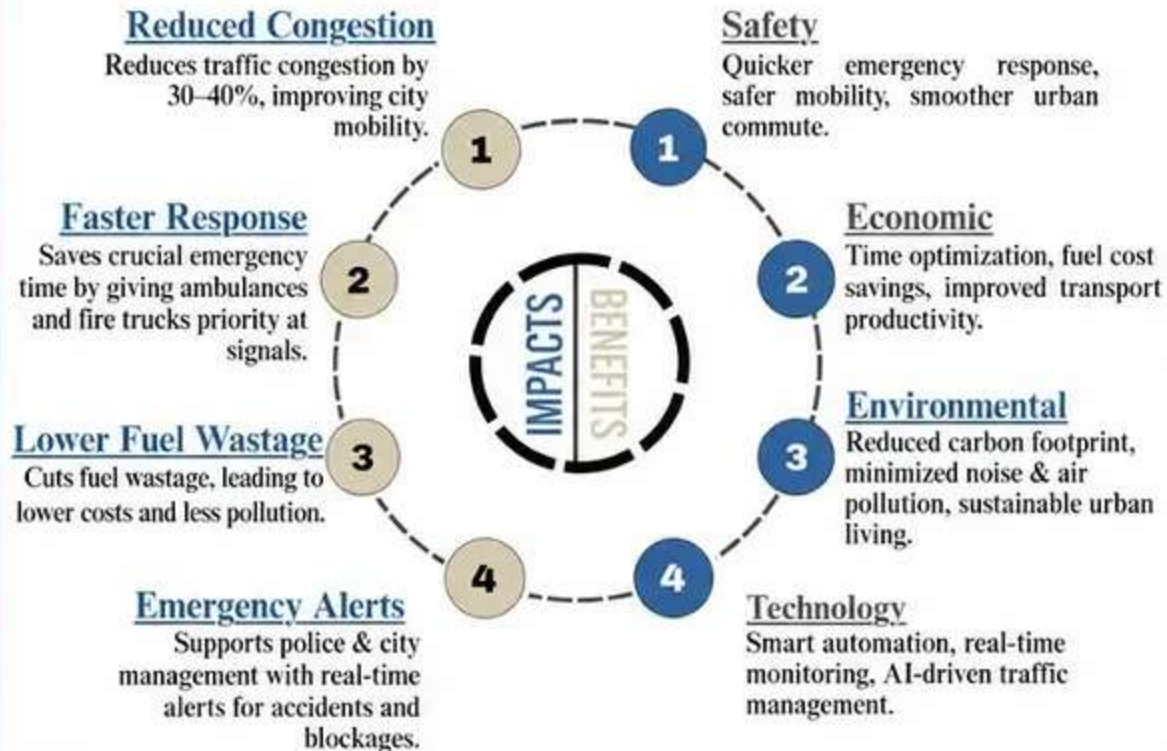




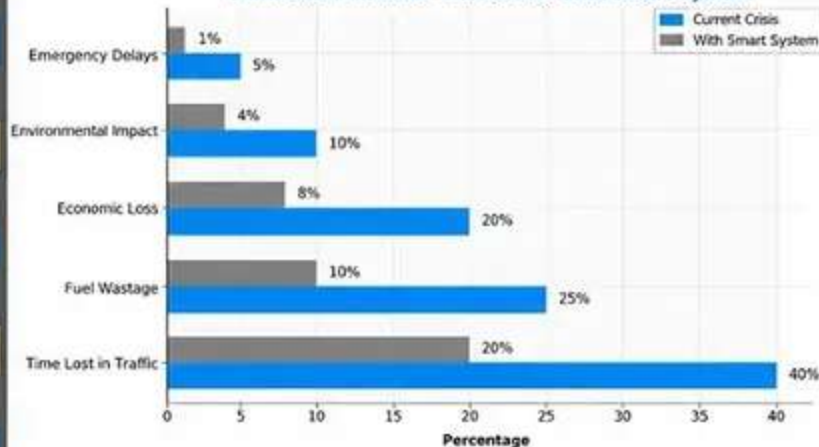
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IMPACT AND BENEFITS



Traffic Crisis in India - Current vs. With Smart System



From Gridlock to
Green Light



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RESEARCH AND REFERENCES



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Adapt signals to
improve flow

Analyze data from
various sources

Algorithms enable
smarter decisions

Inefficient traffic
flow in cities

Improved flow and
reduced congestion

YOLO:
For Real-Time
Vehicle
Detection

[LINK](#)

Omnidim:
Authority
Notification
Agent

[LINK](#)

NITI Aayog:
Smart Cities
and
Infrastructure

[LINK](#)

MoRTH:
Road
Accidents in
India Report

[LINK](#)

IEEE Xplore:
Intelligent
Traffic Signal
Control
Systems

[LINK](#)

Emergency
Vehicle
Preemption
(EVP)

[LINK](#)

Adaptive
Traffic Signal
Control

[LINK](#)

